

Chapter 8 Advanced Counting Techniques

Chapter Summary

- Applications of Recurrence Relations
- Solving Linear Recurrence Relations
 - Homogeneous Recurrence relations
 - Nonhomogeneous Recurrence relations
- Divide-and-Conquer Algorithms and Recurrence relations
- Generating Functions
- Inclusion-Exclusion
- Applications of Inclusion-Exclusion

Section 8.1 Applications of Recurrence Relations

Section Summary

- Applications of Recurrence Relations
- Algorithms and Recurrence Relations

Recurrence Relations

Definition: An equation that a_n can be expressed by previous terms of the sequence.

The sequence that satisfies the recurrence relation is called **a solution**.

The **initial condition** expresses the terms before the recurrence relation takes effect.

Example:

- Rabbit and Fibonacci Numbers: $f_n = f_{n-1} + f_{n-2}; f_1 = 1, f_2 = 1$
- The tower of Hanoi: $H_n = 2H_{n-1} + 1, H_1 = 1$ for one disk. Then $H_n = 2^n - 1$
- Counting bit strings:
Finding the number of bits that have length n and no two consecutive 0.
 $a_n = a_{n-1} + a_{n-2}; a_1 = 2, a_2 = 3$. Similar to Fibonacci. ($a_n = f_{n+2}$)
- Counting the Ways to Parenthesize a Product
 C_n is the number of ways to parenthesize the product of $n + 1$ numbers.
Always one $\cdot$ outside all parentheses. Then it can be between x_k and x_{k+1} . Then
 $C_n = C_0C_{n-1} + \cdots + C_{n-1}C_0 = \sum_{k=0}^{n-1} C_kC_{n-k-1}; C_0 = 1, C_1 = 1$.

Section 8.2 Solving Linear Recurrence Relations

Section Summary

- Linear Homogeneous Recurrence Relations
- Solving Linear Homogeneous Recurrence relations with constant coefficients
- Solving Linear Nonhomogeneous Recurrence relations with constant coefficients.

Linear Homogeneous Recurrence Relations

Definition:

A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ where $c_1, c_2, \dots, c_k$ are real numbers and $c_k \neq 0$.

Linear: right hand is linear combination

homogeneous: all coefficient is constant and all terms is the multiple of a_j .

Degree k : a_n is expressed by previous k terms.

The sequence are unique that $a_0 = C_1, \dots, a_{k-1} = C_k$.

Solving Linear Homogeneous Recurrence Relations

The forms :

$$a_n = r^n$$

where r is a constant.

Thus the characteristic equation:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$$

And find solutions of r . And the solutions are called the characteristic roots of recurrence relation.

Solving Linear Homogeneous Recurrence Relations of Degree Two

Theorem 1:

For $a_n = c_1 a_{n-1} + c_2 a_{n-2}$, suppose the $r^2 - c_1 r - c_2 = 0$ has two distinct roots r_1 and r_2 , then the solution of the sequence is

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$$

where α_1, α_2 are constants.

The Solution when there is a Repeated Root

Theorem 2:

Same condition as theorem 1 but only one repeated root r_0 , then the solution is

$$a_n = (\alpha_1 + \alpha_2 n) r_0^n$$

where α_1, α_2 are constant.

Solving Linear Homogeneous Recurrence Relations of Arbitrary Degree

Theorem 3:

For $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$, if $r^k - c_1 r^{k-1} - \dots - c_k = 0$ has k distinct roots $r_1, r_2, \dots, r_k$ then the solution is

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \dots + \alpha_k r_k^n$$

The General Case with Repeated Roots Allowed

Theorem 4:

Same condition as theorem 3 but t distinct roots $r_1, \dots, r_t$ with multiplicities $m_1, \dots, m_t$ and $m_1 + m_2 + \dots + m_t = k$. Then the solution is

$$\begin{aligned} a_n = & (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ & + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ & + \dots + (\alpha_{t,0} + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

Linear Nonhomogeneous Recurrence Relations with Constant Coefficients

Definition:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n)$$

where $c_1, \dots, c_k$ are constant and $F(n)$ only depends on n and not always zero.

And the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

is called the **associated homogeneous recurrence relation**.

Solving Linear Nonhomogeneous Recurrence Relations with Constant Coefficients

Theorem 5:

For a particular solution $\{a_n^{(p)}\}$ for $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$, then **every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$** where $\{a_n^{(h)}\}$ is the solution of associated homogeneous recurrence relation $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$.

Steps:

- Solve a particular solution depending on the form of $F(n)$
- Solve the solution of associated homogeneous recurrence relation
- Use initial conditions to solve constant

Section 8.3 Divide-and-Conquer Algorithms and Recurrence Relations

Section Summary

- Divide-and-Conquer Algorithms and Recurrence Relations
- Examples
- Master Theorem
- Closest Pair of Points

Divide-and-Conquer Algorithmic Paradigm

Definition:

Divide a problem into one or more instances of the same problem of smaller size and then conquer the problem using the solutions of the smaller problems to get the solutions of original problem.

Divide-and-Conquer Recurrence Relations

- Suppose divide a problem of size n into a subproblems
- Assume each subproblem is of size n/b .
- Suppose $g(n)$ extra operations are needed in the conquer step
- Then the divide-and-conquer recurrence relation is $f(n) = af(n/b) + g(n)$.

Example:

- Binary search

$$f(n) = f(n/2) + 2$$

- Merge sort

$$M(n) = 2M(n/2) + n$$

- Fast Multiplication of Integers

$ab = (2^{2n} + 2^n)A_1B_1 + 2^n(A_1 - A_0)(B_0 - B_1) + (2^n + 1)A_0B_0$. Thus the multiplication of $2n$ -bit integer can be done by three multiplications of n -bit integers with additions, subtractions, shifts

$$\text{Thus } f(2n) = 3f(n) + Cn.$$

Estimating the Size of Divide-and-Conquer Functions

Theorem 1:

$f(n)$ is increasing function and

$$f(n) = af(n/b) + cn^d$$

where n is divisible by b .

then

$$f(n) \text{ is } \begin{cases} O(n^{\log_b a}) & \text{if } a > 1 \\ O(\log n) & \text{if } a = 1 \end{cases}$$

Example: Complexity of Binary Search is $O(\log n)$

Theorem 2:

$f(n)$ is increasing function and

$$f(n) = af(n/b) + cn^d$$

Then

$$f(n) = C_1 n^{\log_b a} + C_2 n^d$$

where $C_1 = \frac{cb^d}{a-b^d} + f(1)$, $C_2 = \frac{cb^d}{b^d-a}$.

where $n = b^k$ and k is a positive integer greater than 1, and c and d are real numbers and c is positive, d is nonnegative.

then

$$f(n) \text{ is } \begin{cases} O(n^d) & \text{if } a < b^d \\ O(n^d \log n) & \text{if } a = b^d \\ O(n^{\log_b a}) & \text{if } a > b^d \end{cases}$$

Example: Complexity of Merge sort is $O(n \log n)$; complexity of fast integer multiplication is $O(n^{\log 3})$.

Section 8.4 Generating Functions

Section Summary

- Generating Functions
- Counting Problems and Generating Functions
- Useful Generating Functions
- Solving Recurrence Relations using Generating functions
- Probing identities using Generating functions

Generating Functions

Definition:

The generating function for sequence $a_0, \dots, a_k, \dots$ of real number is the infinite series

$$G(x) = a_0 + a_1x + \dots + a_kx^k + \dots = \sum_{k=0}^{\infty} a_kx^k$$

Generating Functions for Finite Sequences

The generating function for sequence $a_0, \dots, a_n$ is

$$G(x) = a_0 + a_1x + \dots + a_nx^n$$

Useful Generating Functions

TABLE 1 Useful Generating Functions.

$G(x)$	a_k
$(1+x)^n = \sum_{k=0}^n C(n, k)x^k$ $= 1 + C(n, 1)x + C(n, 2)x^2 + \cdots + x^n$	$C(n, k)$
$(1+ax)^n = \sum_{k=0}^n C(n, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n, 2)a^2 x^2 + \cdots + a^n x^n$	$C(n, k)a^k$
$(1+x^r)^n = \sum_{k=0}^n C(n, k)x^{rk}$ $= 1 + C(n, 1)x^r + C(n, 2)x^{2r} + \cdots + x^{rn}$	$C(n, k/r)$ if $r \mid k$; 0 otherwise
$\frac{1-x^{n+1}}{1-x} = \sum_{k=0}^n x^k = 1 + x + x^2 + \cdots + x^n$	1 if $k \leq n$; 0 otherwise
$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k = 1 + x + x^2 + \cdots$	1
$\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k = 1 + ax + a^2 x^2 + \cdots$	a^k
$\frac{1}{1-x^r} = \sum_{k=0}^{\infty} x^{rk} = 1 + x^r + x^{2r} + \cdots$	1 if $r \mid k$; 0 otherwise
$\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty} (k+1)x^k = 1 + 2x + 3x^2 + \cdots$	$k+1$
$\frac{1}{(1-x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)x^k$ $= 1 + C(n, 1)x + C(n+1, 2)x^2 + \cdots$	$C(n+k-1, k) = C(n+k-1, n-1)$
$\frac{1}{(1+x)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)(-1)^k x^k$ $= 1 - C(n, 1)x + C(n+1, 2)x^2 - \cdots$	$(-1)^k C(n+k-1, k) = (-1)^k C(n+k-1, n-1)$
$\frac{1}{(1-ax)^n} = \sum_{k=0}^{\infty} C(n+k-1, k)a^k x^k$ $= 1 + C(n, 1)ax + C(n+1, 2)a^2 x^2 + \cdots$	$C(n+k-1, k)a^k = C(n+k-1, n-1)a^k$
$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$	$1/k!$
$\ln(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$	$(-1)^{k+1}/k$

Note: The series for the last two generating functions can be found in most calculus books when power series are discussed.

Counting Problems and Generating Functions

For a equation with more unknowns constrained by integers, write the expression that is the product of **generating functions of each unknowns**. Then calculate and obtain the result by corresponding coefficient.

For example, the number of solution of $e_1 + e_2 + e_3 = 17$ where

$2 \leq e_1 \leq 5, 3 \leq e_2 \leq 6, 4 \leq e_3 \leq 7$. Then the expression

$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7)$ and the coefficient of x^{17} is 3 thus it is the answer.

Section 8.5 Inclusion-Exclusion

Section Summary

- The principle of Inclusion-Exclusion
- Examples

Principle of Inclusion-Exclusion

Two Finite Sets

The number of elements in the union of two finite sets:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Three Finite Sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

The Principle of Inclusion-Exclusion

Theorem 1:

Let $A_1, \dots, A_n$ be finite sets, then

$$|A_1 \cup A_2 \cup \dots \cup A_n| = \sum_{1 \leq i \leq n} |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

Section 8.6 Applications of Inclusion-Exclusion

Section Summary

- Counting Onto-Functions
- Derangement

The Number of Onto Functions

Theorem 1:

Let m and n be positive integers with $m \geq n$, then

$$n^m - C(n, 1)(n-1)^m + C(n, 2)(n-2)^m - \cdots + (-1)^{n-1}C(n, n-1) \cdot 1^m$$

onto functions from a set with m elements to a set with n elements.

Derangements

Definition:

A derangement is a **permutation** of objects that leaves **no object in the original position**.

Theorem 2:

The number of derangements of a set with n elements is

$$D_n = n! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots + (-1)^n \frac{1}{n!} \right]$$